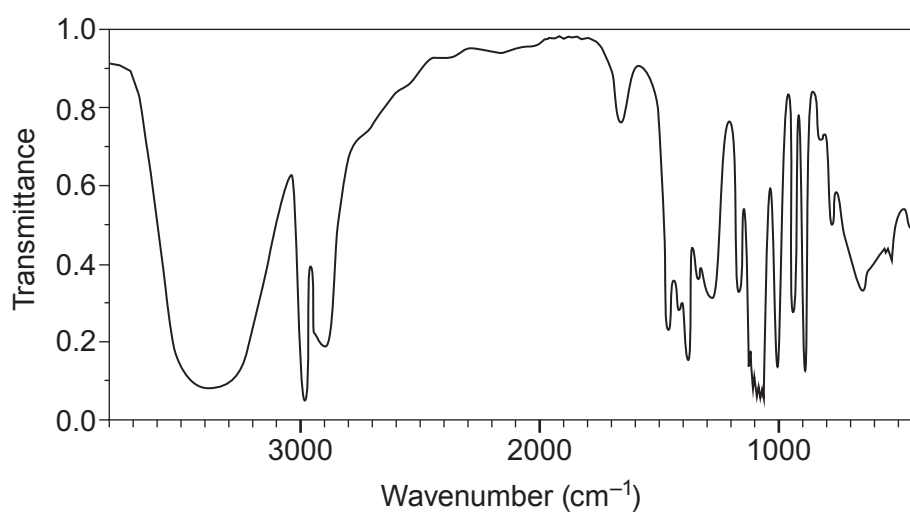
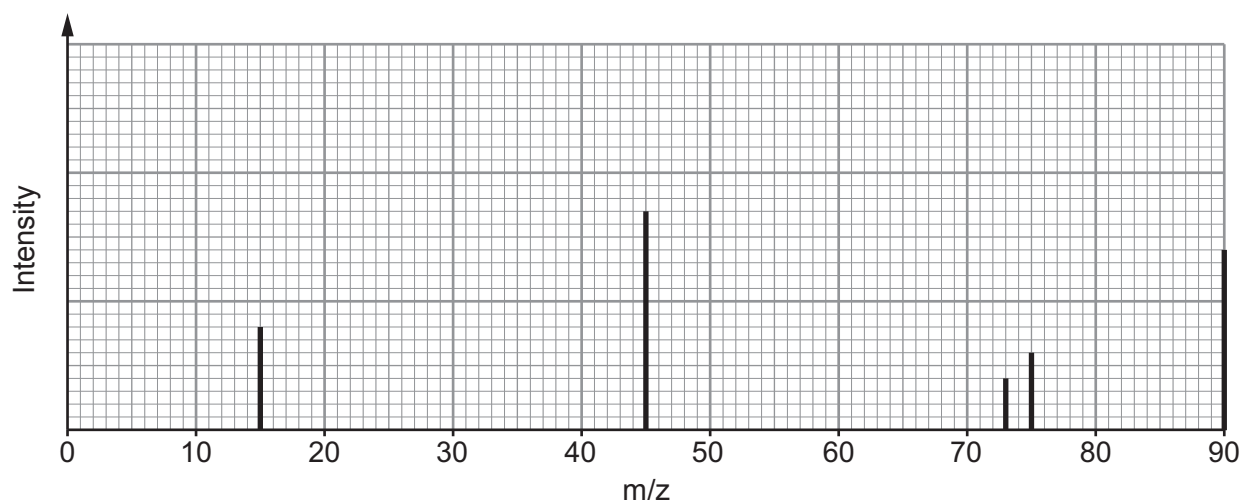


11. Compound **X** contains only carbon, hydrogen and oxygen. On analysis it was found to contain 53.3 % carbon and 11.1 % hydrogen by mass.

A simplified form of the mass spectrum and the infrared absorption spectrum for **X** are shown.



The low resolution  $^1\text{H}$  NMR spectrum of **X** has three peaks.

When **X** is warmed with excess acidified potassium dichromate(VI) there is a colour change. The organic product of this reaction does **not** react with aqueous sodium carbonate.



- (a) Use **all** the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

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Structure of **X**



- (b) (i) State the type of reaction that occurs when **X** is warmed with acidified potassium dichromate(VI). [1]

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- (ii) Draw the structure of the organic product formed when **X** reacts with acidified potassium dichromate(VI). [1]

